

Design Verification Plan & Report - Virtual Test (VBF-T5R1SINGLEMODEL-DVPR-001)

> Case: t5_r1_singlemodel. Virtual test version: every value from simulation output / parameter set, source-annotated. Conditions not simulatable are honestly "N/A (requires physical experiment)".

Item	Condition	Result value	Determination (vs criteria)	Source
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1C discharge capacity	`run-pyamm --protocol 1C_discharge --mode dfn`, 25 °C ambient	5.0374 Ah	informational (no capacity criterion in entry 0)	`cell/r5_final_archN_1c`
Energy density	contract-caliber calc-energy on 1C discharge	533.18 Wh/kg	✓ PASS vs ≥ 500.94 Wh/kg (+32.2)	`cell/r5_final_archN_energy_dfn.json:energy_density_wh_k`
4C fast-charge temperature rise	`run-pyamm --protocol 4C_charge_45C --thermal lumped --mode dfn`, 45 °C ambient	T_max 330.50 K (57.35 °C)	✓ PASS vs ≤ 333.15 K (2)	
4C fast-charge lithium plating	same run, `--plating`-caliber anode potential series	anode potential min +0.0215 V (never < 0 V)	✓ PASS vs plated = false (+21.5 mV marg	
Voltage window	parameter set	2.5 - 4.2 V	informational	`OKane2022:Lower/Upper voltage cut-off [V]`
Nail penetration	—	N/A (beyond pure-simulation boundary, requires physical experiment)	N/A	—
Overcharge to thermal runaway	—	N/A (thermal-runaway ODE module available in library but not part of this task's criteria)	N/A	—
Crush / drop	—	N/A (requires physical experiment)	N/A	—
Cycle life	—	N/A (no aging run in this case; aging protocol exists in library)	N/A	—
Rate-pulse (HPPC) internal resistance	—	N/A (1C-derived DCR 15.89 mΩ reported as informational in datasheet)	N/A	`cell/r5_final_archN_energy_dfn.json:dcr_ohm`

Conclusion

****PASS**** — all three entry-0 criteria verified by simulation: ED 533.18 Wh/kg ≥ 500.94 ✓; T_max 330.50 K ≤ 333.15 K ✓; plated = false ✓ (evidence entries in log.jsonl R4/R5 evaluate, mechanically generated by `bda log-evaluate`).

Uncovered items (honest limitations for the paper): nail penetration, overcharge-to-thermal-runaway, crush, drop, cycle life, rate-pulse DCR — all outside the pure-simulation scope of this case; plus electrolyte-additive cell-level effect (no additive bridge parameter) and true-compute molecular endorsement (real_compute = false).